5. ARES: The Mars Port (Phase 4)

Mars atmosphere is 0.6% of Earth's. The hard parts of the Earth machine are nearly free there.

The tube holds back 0.6% of the pressure the Earth tube fights; the plasma window's job is 99.4% done by the planet; exit heating is a fraction of the Earth case. Dust storms, not aerodynamics, are the main environmental design driver.

```
Mars tracks from L = v²/2a, human-rated a = 3.3 g = 32.4 m/s²:
Low Mars orbit (3.4 km/s): L = 3,400² / (2 × 32.4) = 178 km, t = v/a = 105 s
Mars escape (5.0 km/s): L = 5,000² / (2 × 32.4) = 386 km, t = 154 s
Energy: E = ½v² ⇒ orbit ½ × 3,400² = 5.8 MJ/kg = 1.6 kWh/kg; escape ½ × 5,000² = 12.5 MJ/kg = 3.5 kWh/kg
```

Element	Figure	Notes
Orbit track	~3.4 km/s → ~180 km @ 3.3 g	105-second run to low Mars orbit
Escape track	~5.0 km/s → ~390 km @ 3.3 g	155-second run; between Moon and Earth in scale, far easier than Earth in engineering
Energy	1.6 / 3.5 kWh per kg	orbit / escape
Staging	Phobos & Deimos	Catch points and propellant depots, the warehouse district; SEP tugs fly the cruise

Every Mars mission arrives needing return propellant. ISRU solves it on the surface (Sabatier methalox from CO_2 and water ice); ARES solves it in orbit, propellant rides to Phobos at electromagnetic cost and waits in a depot for any departing vehicle. Return cargo, manufactured exports, and samples ride the same track. **ARES makes Mars a place you can leave as easily as you arrive.**

> **The Moon as deep-space launch platform.** A Mars-transfer from the lunar surface costs ~3.0 km/s (≈1.25 kWh/kg), roughly one-fifteenth the energy of the same trip from Earth's surface. Anything Mars-bound that can be built on, or staged through, the Moon ships from the lunar track: relay satellites, sensor networks, pre-positioned supplies. Starship carries crew and cargo that must arrive fresh; the mass drivers send everything else ahead, early and cheap. The Mars supply chain runs through the Moon, not around it. ARES is Phase 4: it inherits a mature vacuum-world kit (SELENE, Phase 1; ASTERIA, Phase 2–3) and proven atmosphere subsystems (Chimborazo, Phase 2–3).

6. The Build Order

The lightning flashes before the thunder is heard: SELENE launches silently in vacuum before BRONTE shakes the air on Earth.

Phase	System	What gets built	What it proves / earns
1	**SELENE PoC** (Moon)	1–3 km surface track, 50–100 g, south-pole ridge	2.9 km @ 100 g clears lunar escape; exports water/regolith/metal to cislunar buyers from day one; grows to 10–12 km (30 g) then ~50 km (3 g)
2–3	**Chimborazo** (Earth) + **ASTERIA**	14 km first build of the straight through-mountain line (tube, plasma window, aero-shell) running in parallel with SELENE-class kits stamped on Phobos, Deimos, Ceres, the Belt	Chimborazo earns DoD test revenue and proves cost-per-km; ASTERIA deploys proven vacuum-world technology across airless bodies, both run concurrently, the core bet and the Earth demonstrator advance together
4	**ARES** (Mars)	~180 km track, featherweight tube	Mars becomes a port; the freight network closes
5	**BRONTE full** (Earth)	Extend Chimborazo west to ~21.3 km straight (Mach 11.2); beyond that, the 1,000 km-class 3 g injector if economics prove out	The endgame, built only on validated cost data

SELENE is ~300× shorter than any full Earth injector and skips every atmosphere subsystem. Chimborazo Phase 1 is ~70× shorter and proves the hard atmosphere parts before any large commitment. They de-risk in parallel.

7. How the Network Moves Mass

Four launchers, one supply chain. Once they exist, moving mass around the inner system mostly costs what running the launchers costs.

- **Earth sends what only Earth can build**, machines, electronics, medicine. The eventual full injector runs 80 t/day (29,000 t/yr); the Chimborazo demonstrator's job is to earn that machine the right to exist. (Rev 1 wrongly pinned the 80 t/day cadence on the demonstrator itself.)
- **The Moon sends raw mass**, propellant feedstock, shielding regolith, structural metal, fired at 2.4 km/s toward cislunar catch points. At full cadence the cislunar propellant supply stops being constrained by what Earth can lift.
- **Mars closes the return trip**, ISRU methalox, samples, and exports ride ARES to Phobos/Deimos depots on electricity.
- **SEP tugs connect the nodes**, weeks per leg, nearly propellant-free; chemical engines do only final descents. The launchers do all the heavy lifting off every surface.
- **Each node funds the next.** Rockets deliver SELENE's first coils (the last rocket buy the lunar build needs); SELENE exports fund Chimborazo; Chimborazo revenue and data fund ASTERIA; ASTERIA builds ARES. Every airless body added makes the whole network richer.

8. The Skyhook: How Cost Falls Toward $5/kg

A rotating tether catches pods and flings them higher. No propellant. More payload per launch.

A long tether rotates in low orbit; its bottom tip sweeps backward, moving slower than orbital velocity. A pod that arrives at the tip gets grabbed; half a rotation later it is released up high, moving faster, to the Moon, GTO, or wherever the mission calls. With a 1.5 km/s tip speed, the launcher needs 6.3 km/s instead of 7.8: ~35% less energy, a slower exit (less heating), and the kick stage shrinks from ~20% of pod mass to 2–5% for trim, the freed 15–18% becomes payload. That shift is where $5/kg comes from.

- **Keeping it up:** electrodynamic reboost (current through the tether against Earth's field, demonstrated physics, no propellant) plus momentum returned by catching descending capsules.
- **The lunar extension:** a cislunar rotovator catches SELENE pellets and redirects them to LLO, L1, or Earth-transfer; an Earth-orbit tether receives. Cargo crosses cislunar space without burning its own propellant at all.
- **Provenance:** momentum-exchange tethers date to the 1970s; the Boeing/Tethers Unlimited HASTOL study (NASA NIAC, 1999–2001) detailed a 600 km tether with 3.5 km/s tip speed catching Mach-10 aircraft-launched payloads. Materials and manufacturing at scale, not new physics.

9. If the Full Earth Injector Gets Built

Not the plan, ASTERIA is the plan. But if ASTERIA works, this is what becomes possible.

A full Earth injector, ultimately the 1,000 km-class, 3 g, 8 km/s human-rated machine, is the most consequential infrastructure ever proposed. The key point: if SELENE works in vacuum and Chimborazo validates the atmosphere subsystems and the cost-per-kilometer, the full injector stops being a physics question and becomes a construction question.

What $5–20/kg to orbit changes

- **Satellites:** launch stops dominating cost, a 2,000 kg satellite rides for ~$20K at $10/kg vs the $83–97M a GPS III mission actually paid. Constellations replace on maintenance schedules.
- **Stations:** ISS-class logistics (~$3.1B/yr, ~$1.8B of it transport) collapses ~300×; thousand-person stations become economically normal.
- **Orbital manufacturing:** ZBLAN fiber, crystals, alloys, Flawless Photonics drew ~12 km of ZBLAN on ISS in 2024 (1,141 m in one day vs a 25 m prior record); the candid caveat is that published proof of reproducibly superior fiber is still pending. The queue is real either way, and it is waiting on freight prices.
- **Deep space:** with cheap Earth-to-orbit plus lunar propellant, Mars missions stop being nation-sized budget exercises; the Belt becomes commercially reachable.

> **The strategic picture.** The operator of an 80 t/day injector controls the access ramp to space, 29,000 t/yr can replenish a damaged constellation in hours. The host nation stops paying for space access and starts charging for it. ASTERIA pays for itself; the Earth injector is what ASTERIA eventually makes possible.

10. Theoretical Scaling

Pure forward-looking theory. None of this is in the build plan, it is what the architecture could become if the demonstrator succeeds and the enabling technology keeps maturing.

> **Read this section as speculation, not specification.** Everything below assumes the Phase-1 demonstrator has already worked and that several technologies have advanced past where they sit in 2026. These are not designs. They are the shape of the ceiling, included because the question "how far does this scale?" deserves an honest, numbered answer rather than hand-waving.

10.1 BRONTE (Chimborazo, Earth): bore, pod, and power scaling

> **The one rule that governs all BRONTE scaling: bore sets the chunk size, power sets the tonnage.** The energy to place a kilogram in orbit is fixed by physics; a wider tube does not make a tonne cheaper, it makes each shot bigger and rarer for the same megawatts. More annual tonnage comes only from more installed power. Bigger bores and more barrels change *how* you deliver the mass; the power plant decides *how much*.

Theoretical A, the Ø5 m bore

Payload scales with bore cross-section (diameter^2): widening the Ø3.2 m payload bay to Ø5 m is a **2.44× larger payload per shot**. The same exit velocity is required, so energy per shot scales with it.

What would have to advance first: the plasma window, already the program's hardest R&D at Ø4 m, would need to hold a still-larger aperture; the drive ring would need more REBCO at a better cost per kiloamp-metre than 2026 production; tunnelling would need bores beyond today's ~17.6 m TBM record; and trackside storage would need to swallow ~80 MWh per shot. Each is an extrapolation of a real 2026 capability, not a new invention, but all four have to arrive together.

Theoretical B, the multi-barrel battery

Instead of one fat tube, place several independent barrels side by side on one site, ~20 m apart through the rock (a rock pillar between Ø12 m bores, the way twin- and triple-bore road tunnels are already built), 5–10 m apart on open viaduct. They share one tunnel campaign, one substation, one cryoplant, one vacuum plant, one control centre, and, critically, **one storage farm fired round-robin:** while barrel 1 recharges, barrel 2 fires, then 3, so the grid sees a steady draw instead of three spikes and the site fires far more often than any single barrel.

What this does and does not buy: it is mostly engineering scale-up, not new physics, but it is gated on the demonstrator first proving the per-barrel cost, and on building power to match. Three barrels deliver ~3× the tonnage for under 2× the capital, because the expensive infrastructure (and the permitting fight) is paid once. It does not escape the power rule: three barrels at full cadence draw three times the megawatts.

Theoretical C, a longer pod

The baseline ASTRAPE is 23 m. The pod can be made far longer before physics objects: with the full-hull distributed drive there is no tail to crush (axial buckling only becomes a hard wall past ~140 m of carbon structure at 30 g, ~174 m for the wider Ø5 m hull). With BRONTE now a straight line (no terminal bend, §4), the old ~280 m bend-clearance limit is gone, column buckling is the only ceiling. Every length below sits well under it, ordinary rigid structure, no articulation needed; even at 70 m the binding constraint is *handling and loading* a single object that long, not strength. Because the drive force grows with engaged length, a longer pod can be proportionally heavier at the same g, so payload scales roughly with length × bore cross-section.

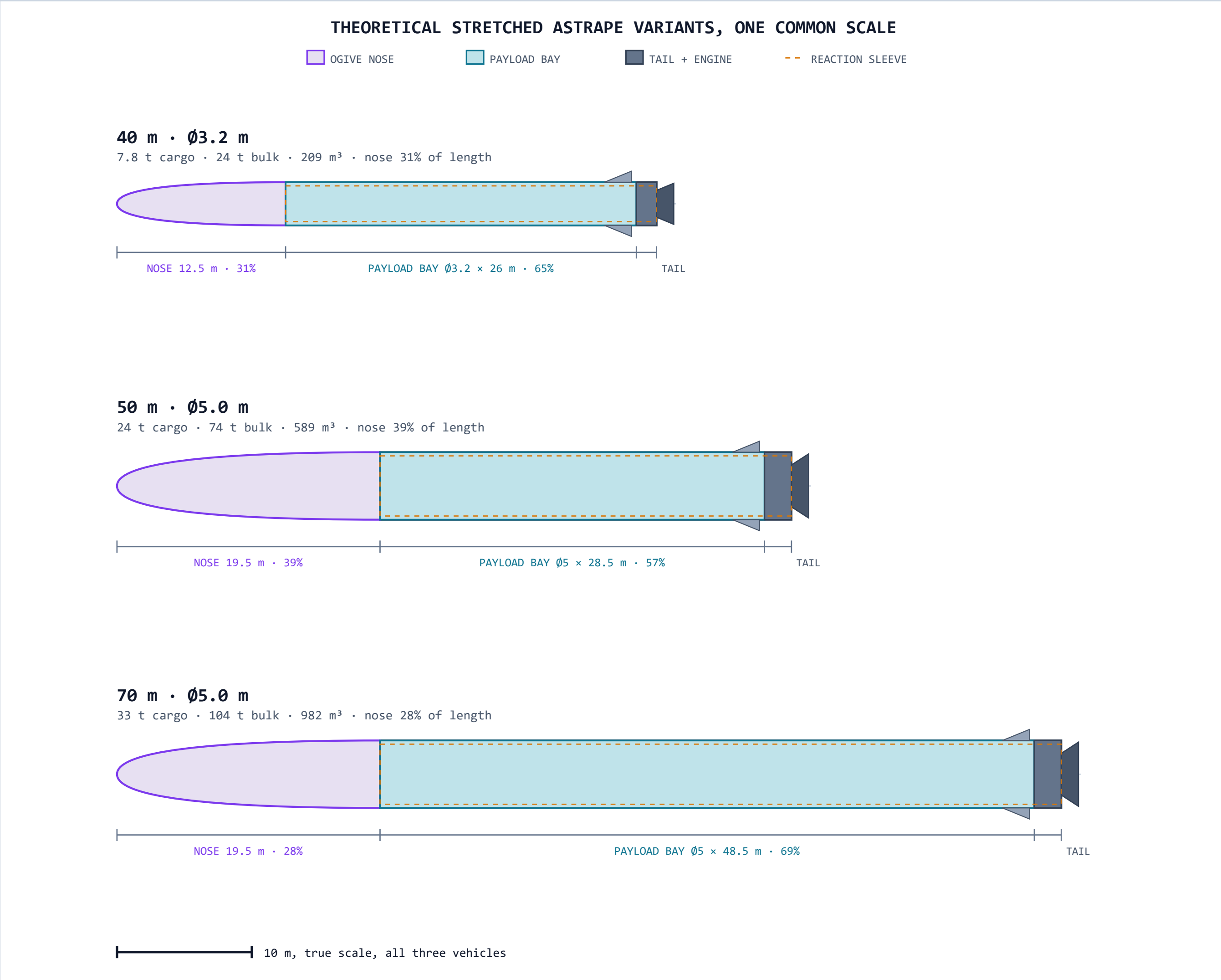
Pod × bore	Gross mass	Cargo → orbit (30 g)	Bulk → orbit (50–77 g)	Bay volume
23 m · Ø3.2 m (baseline)	~20 t	4.5 t	14 t	~64 m^3
30 m · Ø3.2 m	~26 t	~6 t	~18 t	~129 m^3
40 m · Ø3.2 m	~35 t	~7.8 t	~24 t	~209 m^3
50 m · Ø5 m (theoretical)	~106 t	~24 t	~74 t	~589 m^3
70 m · Ø5 m (theoretical · pushing the limit)	~149 t	~33 t	**~104 t**	~982 m^3

A fat bore only pays off with a long pod, at Ø5 m the nose cone alone is ~20 m, which is why the theoretical Ø5 m variants are drawn long. The 50 m · Ø5 m pod already approaches one Starship's per-launch mass and volume (~100 t / ~1,000 m^3); the 70 m · Ø5 m is about the longest single rigid pod worth building, ~104 t of bulk in one shot, ~982 m^3, beyond which you would articulate the pod or accept the handling penalty.

> **"Bulk to orbit" is not fire-and-forget.** Even inert bulk cargo (propellant, water, metal) still needs **a small kick stage** to circularize at apogee, the gun raises the apoapsis, but something must add the last ~0.85 km/s to turn a ballistic arc into a stable orbit, plus **guidance, RCS thrusters, and attitude control** to steer, hold orientation, and rendezvous with the orbital catch-point or depot. The bulk-mode payload figures above are *after* that kick stage and avionics are subtracted from gross mass. Nothing reaches a usable orbit on the launcher alone; the launcher just does ~90% of the work for ~1% of the propellant.

Theoretical D, what the long pods actually look like

Three theoretical stretched ASTRAPE variants, drawn to one common scale so length and bore compare directly. The ogive nose is fixed by aerodynamics at ~3.9× the diameter, so a fatter bore spends more of the pod on nose cone before any payload bay begins, which is exactly why a wide bore only earns its keep on a long pod.



Nose cone is a ~3.9:1 ogive (length ≈ 3.9 × diameter): ~12.5 m at Ø3.2 m, ~19.5 m at Ø5 m, a fixed overhead that a longer body amortizes. At Ø5 m the 50 m pod is ~39% nose; stretch it to 70 m and the nose share drops to ~28% while the payload bay nearly doubles (28.5 m → 48.5 m). The skinny 40 m · Ø3.2 m is already ~65% payload bay. All dimensions theoretical · © Perpetual Technologies.

Theoretical E, the price of scale: power and cost

Energy per shot is fixed by physics: ½ × launched mass × exit-speed². A heavier pod, or a faster (bulk-mode) exit, costs proportionally more. Track electricity below at $0.06/kWh; cargo mode exits at Mach 11.2 (3.54 km/s, the straight-line full-line exit) and finishes on a kick stage, bulk mode is driven all the way to orbital speed (7.8 km/s) on the track itself.

Variant (max gross)	Cargo shot, Mach 11.2	Bulk shot, orbital 7.8 km/s
23 m · Ø3.2 m (20 t) (baseline)	~35 MWh · ~$2,100	~169 MWh · ~$10,100
40 m · Ø3.2 m (35 t)	~62 MWh · ~$3,700	~296 MWh · ~$17,700
50 m · Ø5 m (106 t)	~184 MWh · ~$11,100	~896 MWh · ~$53,700
70 m · Ø5 m (149 t)	~259 MWh · ~$15,500	~1,259 MWh · ~$75,500

The cost-per-kilogram barely moves with size, ~$0.58–0.59/kg of electricity in cargo mode, ~$0.72–0.74/kg in bulk mode (bulk does all the work electrically instead of offloading it to a kick stage). That flatness is the whole thesis: the launcher's marginal cost is electricity, and electricity scales linearly. What grows is the *capital*, the storage, the power feed, and the structure.

Where the power comes from. The trick is two stages: a steady generator slowly fills an energy store, and the store dumps it in seconds. The grid (or power plant) never sees the peak, only the trackside flywheel/SMES farm does. Peak draw at exit is force × speed: the baseline shot peaks at ~21 GW, the 70 m cargo shot at ~160 GW, and the 70 m *bulk* shot at roughly ~700 GW, on the order of the entire United States grid, sourced for ~13 seconds from stored energy, then gone. Only pulsed-power kit (flywheels, superconducting magnetic storage, compulsators) can deliver that; chemical batteries physically cannot.

To get a feel for it: one 70 m bulk shot needs ~1,259 MWh in the store, about what a 1 GW power station makes in 75 minutes, released in ~13 seconds (≈40% of the energy in the world's largest grid battery, Moss Landing, discharged far faster than that battery could ever manage). Firing ~8 such shots a day means roughly **~420 MW of generation feeding that one barrel around the clock**, a dedicated mid-size power station: one nuclear unit, a ~2 km² solar-plus-storage field, or a major hydro tap (a natural fit for Ecuador's grid). The realistic cargo-mode shot is far gentler: ~259 MWh, ~90 MW of sustained generation, ~155 GW peak. The point stands either way: **the launcher is cheap to fire and expensive to power-build**, pennies of electricity per kilogram, but a serious generation-plus-storage plant behind every barrel.

Theoretical F, why the track has no curve

BRONTE is a single straight line at 12.3°, and it is worth showing *why* we never add even a gentle terminal curve to steepen the exit: at these speeds any curve crushes the payload sideways. Centripetal load is $a = v^2/R$, set by speed not mass, so even a very large radius bites hard.

```
If the track curved at radius R = 100 km (a gentle curve at this scale):
cargo 3,540 m/s: a_lat = 3,540² / 100,000 = 125 m/s² = ~13 g sideways
bulk 7,800 m/s: a_lat = 7,800² / 100,000 = 608 m/s² = ~62 g sideways
Straight track (R → ∞): a_lat = v²/R → 0 g. The only load is axial, 30 g cargo, 3 g human.
```

Load case	If it curved (R = 100 km)	Straight (the design)
Lateral g, cargo exit	~13 g	**0 g**
Lateral g, bulk exit	~62 g	**0 g**
Peak ring side-force, 70 m bulk pod	~91 MN (29× baseline)	**~0**, levitation/centering only
Axial launch g	30 g cargo / 3 g human	30 g cargo / 3 g human

So the straight line is not a compromise, it is the only geometry that keeps the payload within its g-limit at exit speed. The 360° ring therefore never has to restrain tens of meganewtons of curve side-thrust, and there is no point where the track fights to *turn* the pod. The pod still carries enormous kinetic energy (set by mass and speed, not by track shape):

```
E_kinetic = ½mv², e.g. 70 m bulk pod: ½ × 149,000 kg × (7,800 m/s)² = 4.53 × 10¹² J
÷ 4.184 × 10⁹ J per tonne TNT = ~1,080 t TNT (≈ 1 kiloton)
```

On a straight track a loss of drive simply lets the pod **coast straight**, the benign failure mode, instead of being flung sideways through the tunnel wall. Containment and an empty downrange corridor still matter; the straight geometry just removes the lateral-g hazard entirely. The only price is a little top speed, a straight 12.3° grade is shorter than a shallow-grade-plus-curve would be (Mach ~11.2 full line), a trade well worth making.

Theoretical G, banking the pulse: how fusion stores its power, and why magnets fit

The peak-power figures above, hundreds of gigawatts for ~13 seconds, are never drawn from the grid. They come from energy banked between shots and released in one burst. That is precisely the problem the fusion and electromagnetic-launch communities have already solved at gigajoule scale, so KERAUNOS would borrow their hardware rather than invent it.

How fusion already does it. Magnetic fusion faces the identical surge problem, its field coils need a pulse the grid cannot give. JET drives its magnets from two flywheel generator-converters that bank gigajoule-scale energy and deliver up to **2,600 MJ per pulse at 100 kA**, supplementing a 575 MW grid feed, and have done so reliably for ~85,000 pulses since 1983. Inertial fusion takes the other route: NIF stores ~**400 MJ per shot** across 3,840 capacitors, the world's most energetic capacitor bank, charging over ~60 seconds and dumping it in a 400-microsecond burst. Two proven templates: spin it up (flywheel) or charge it up (capacitor).

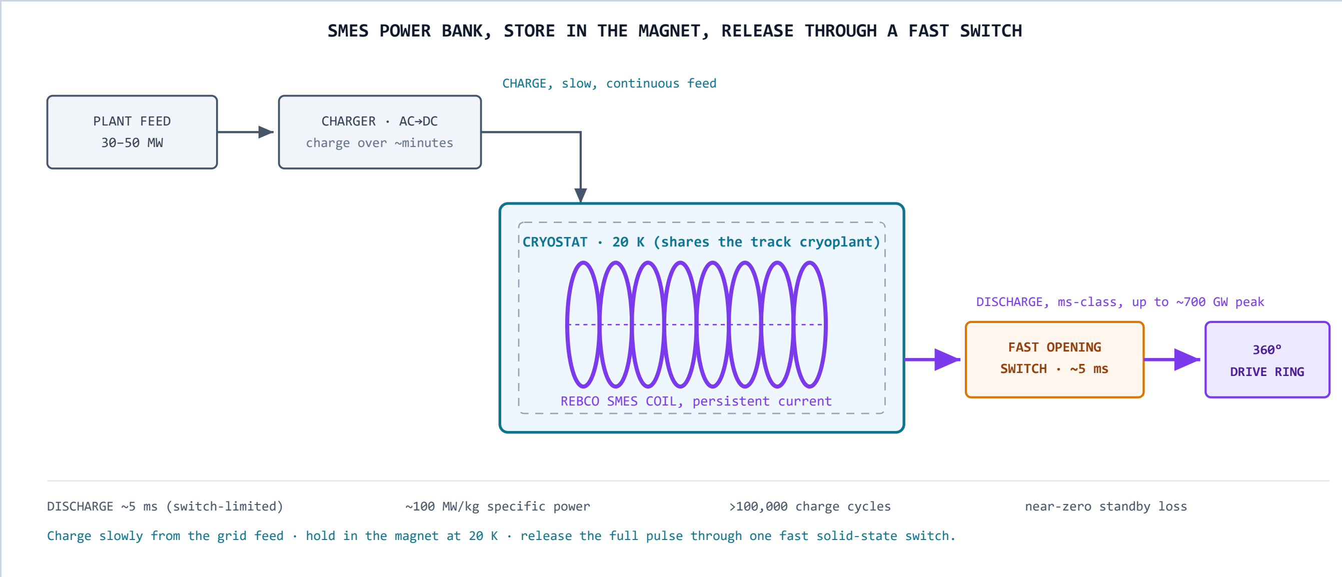
How you bank a pulse	Stores energy as	Power density	Discharge	Proven by
Flywheel / compulsator	rotation (kinetic)	high	seconds	JET; US Navy EMALS (484 MJ)
Capacitor bank	electrostatic field	very high	µs–ms	NIF
SMES, the magnet route	**magnetic field**	**extreme (~100 MW/kg)**	ms · switch-limited	grid stabilizers; lab-scale HTS

The magnet route, and why it fits KERAUNOS best. Superconducting magnetic energy storage (SMES) keeps the energy as current circulating in a short-circuited superconducting coil at near-zero loss; to fire, you open a fast switch and the magnetic field collapses into the load. Its specific power has essentially no ceiling, figures around 100 MW/kg are quoted, which is exactly what a multi-gigawatt pulse wants. It is also fast: a SMES coil goes from full charge to full discharge in roughly **5 milliseconds**, set by the switch rather than the magnet, squarely in the millisecond-to-second band between capacitors (sub-millisecond) and batteries (minutes). And it is uniquely natural here: KERAUNOS is *already* a giant superconducting machine, a 20 K cryoplant, REBCO coils, and a 360° drive ring that is itself a huge inductor. The same cryogenics and the same REBCO supply chain that run the track can run the store, and the drive ring is already holding magnetic energy on every shot. Banking the *next* shot the same way simply closes the loop.

The honest catch, the switch. Inductive storage lives or dies on the opening switch: to release the pulse you must interrupt enormous currents, 100 kA and beyond, in milliseconds, and that switch has historically been the hardest component (early systems used explosively-driven switches; fast solid-state and plasma switches are the modern bet). Rotating compulsators, the preferred rapid-fire railgun supply, sidestep the opening switch entirely, at the cost of being heavy spinning machines. The realistic KERAUNOS answer is a hybrid: proven flywheels or compulsators for the bulk energy, with SMES and a fast solid-state switch for the final high-power shaping, all sharing the launcher's cryoplant. It is the storage farm, not the tube, that most needs invention.

Theoretical H, the power bank and the cables that feed it

A sketch of the magnet store from Theoretical G, and the conductors that move the energy. The design is deliberately asymmetric: the bank is charged *slowly* from a modest feed over minutes, then dumped through a fast switch in milliseconds. Cable sizes below are computed for the current Ø3.2 m line and for the longer 40 m · Ø3.2 m variant; the Ø5 m bore is left for later.



Conductor · where it runs	Current version (~20 t)	Longer 40 m · Ø3.2 m (theoretical)
HV charging feed, full tunnel length, taps each storage station (34.5 kV, continuous)	~30 MW → ~530 A → **3 × 240 mm² Cu**	~50 MW → ~880 A → **3 × 400 mm² Cu**
Pulse bus, storage station → drive coils (REBCO, 20 K, ~10 kV drive bus)	~23 GW → ~2.3 MA → **~5,800 mm²** (≈Ø86 mm); ~240 mm²/sector	~41 GW → ~4.1 MA → **~10,200 mm²** (≈Ø114 mm); ~425 mm²/sector
...the same pulse bus as plain copper (why it must be superconducting)	≈Ø19 cm solid busbar	≈Ø26 cm solid busbar

The striking part: the **grid-side feed is small**, ordinary medium-voltage power cable (a 240 mm² Cu conductor carries ~530 A; reference: an 11 kV 185 mm² XLPE cable is rated ~430 A), because the storage buffers the pulse and the grid only sees the slow recharge. The **pulse bus is enormous in current** but compact and lossless as REBCO (engineering current density ~400 A/mm²), and it shares the launcher's cryoplant; as plain copper the same conductor would be a ~20–26 cm bar. Assumes a 34.5 kV feed and a 10 kV drive bus, a higher drive voltage shrinks the pulse bus proportionally. Routing is shown on the tunnel section (Annex A, Panel D): feed and pulse bus ride the invert service ducts alongside the 20 K / 80 K cryo headers. Ø5 m bore deferred.

How the numbers scale (theoretical)

Configuration	Payload / shot (cargo · bulk)	Energy / shot	Power feed	Annual / site (cargo · bulk)
Ø3.2 m single barrel (demonstrator-derived baseline)	4.5 t · 14 t	~33 MWh	~30 MW	~13,000 t · ~40,000 t
Ø5 m single barrel (theoretical)	~11 t · ~34 t	~80 MWh	~73 MW	~32,000 t · ~99,000 t
Triple Ø3.2 m battery (theoretical)	4.5 t · 14 t (×3, staggered)	~33 MWh ×3	~90 MW	~39,000 t · ~120,000 t

Cargo = 30 g, g-hardened satellites, with a kick stage to orbit. Bulk = 50–77 g inert mass (propellant, water, metal) into a depot. Annual figures assume ~8 effective shots/day per barrel at ~80% availability and that power is scaled to suit.

What it would take to reach 100,000 t/yr to LEO

Approach	Units needed	Notes
SpaceX Starship (reference)	~40 ships	at ~25 flights/ship/yr (Falcon-9-class reuse); ~1,000 flights/yr, ~2.7/day; ~4.6 Mt of methalox/yr
Ø3.2 m barrels	~8	eight sites, or fewer with batteries
Ø5 m barrels (theoretical)	~3	each fed ~73 MW
Triple Ø3.2 m batteries (theoretical)	~3 sites (9 barrels)	three mountains instead of eight

For scale: the entire world placed roughly 1,900 t in orbit in 2024. Every figure in this section is two orders of magnitude past that, and bounded, in the end, not by the launchers but by how much power and how many orbital catch-points can be built to absorb the mass.

PURE THEORETICAL BRONTE SCALING, REQUIRES TECHNOLOGY MATURATION BEYOND 2026, NOT A BUILD PLAN, NOT FOR CONSTRUCTION

10.2 SELENE (Moon): track length and solar power scaling

SELENE scales on completely different axes from BRONTE. There is no bore limit, no plasma window, no tunnel, no underground rock mechanics. The track lays on regolith. The scaling dials are (1) track length, which sets exit speed, and (2) installed solar power, which sets annual tonnage.

Baseline payload bore: Ø3 m. The Ø3 m bore is fixed across all five phases, same stator ring and bucket geometry throughout. The Ø2.5 m × 2.9 m payload bay (14.1 m^3) holds 7,069 kg of bulk at 500 kg/m^3. REBCO ring mass scales with circumference, not bore area, so the full 2.9 km Phase 1b track needs only 6.7 t of superconductor, co-manifested on a single Starship lander mission at roughly $13M delivery cost. The Ø3 m bore enables large packaged cargo, equipment modules, and pressurised tanks from day one, at only $152M additional tape cost over the minimum viable Ø1 m design, 5% of total build budget.

Return missions from Phase 1b. The pod carries a PICA heat shield that doubles as a dust dome at launch and an entry shield at Earth re-entry. Three mission types require no track redesign: (1) **direct Earth atmospheric entry** for samples and high-value cargo (He-3, ZBLAN fiber, pharmaceutical crystals), arriving at ~11 km/s, same class as Apollo CM; (2) **satellite dispensers**, carrying up to ~7 t of hardened satellites per shot (by volume ~15–20 × 100 kg ESPA-class smallsats, or one 3–5 t bus) that each apply their own onboard Δv after release, ~200 m/s for a lunar orbit or L1/L2 halo and ~800 m/s for a GEO transfer (Earth LEO needs a dedicated ~3,100 m/s kick stage); (3) **cislunar depot supply**, where a 150–250 m/s correction burn after escape targets an L1 or L2 propellant depot directly. The Ø3 m bore makes the SELENE track a two-directional logistics node from first operation.

> **The one rule that governs all SELENE scaling: solar power sets the tonnage; track length sets the destination.** Energy per kilogram is fixed by exit speed. More panels = more shots = more mass exported. A longer track at the same g-load delivers the same tonnage per megawatt as a shorter one if the exit speed is the same, but it enables gentler g-loads for crew or fragile cargo.

```
SELENE throughput rule (escape speed, 84% wall-plug efficiency, 90% south-pole illumination):
Wall-plug energy: E = ½v² / η = ½ × 2,385² / 0.84 = 3.384 × 10⁶ J/kg = 0.940 kWh/kg
Annual solar energy: P × 8,760 h × 0.90 = P × 7,884 MWh per installed MW
Annual throughput: 7,884 MWh/MW ÷ 0.940 kWh/kg = ~8,390 t per MW of solar installed per year
Phase 3 direct cislunar (4.2 km/s): E = ½ × 4,200² / 0.84 = 10.5 MJ/kg = 2.917 kWh/kg ⇒ ~2,700
t/MW·yr
```

Solar array	Escape-speed throughput (phases 1b/2/5)	Direct cislunar, 4.2 km/s (phase 3)	Context
8.7 MW (SJSU Phase 1 baseline)	~73,000 t/yr	~24,000 t/yr	38× global 2024 orbital mass, from a 2.9 km track
10 MW	~84,000 t/yr	~27,000 t/yr	comparable to a small photovoltaic farm
50 MW	~420,000 t/yr	~135,000 t/yr	supplies a serious cislunar propellant depot
100 MW	~840,000 t/yr	~270,000 t/yr	closes the cislunar economy independently

South-pole illumination 90% annual average (Gläser et al. 2014, LRO/LOLA). Wall-plug efficiency 84% (see §13). Escape speed 2.385 km/s. Phase 3 exit 4.2 km/s (30 km, 30 g). Figures are peak capacity; actual throughput depends on payload loading cadence.

Track length is the exit-speed dial, not the tonnage dial. The five phases (detailed in §1) cover 2.9 km to 96 km. Every phase uses the same REBCO stator ring and cryo circuit, only the track is longer. There is no bore, no plasma window, no tunnelling machine. The SJSU Phase 1b machine at 8.7 MW already outperforms all of 2024 human spaceflight by volume, from a 2.9 km aluminium-on-regolith track. The SELENE scaling story is simply: more solar panels, same track.

PURE THEORETICAL SELENE SCALING, REQUIRES SOUTH-POLE SITE DEVELOPMENT, NOT A BUILD PLAN

11. Zero New Physics

Every subsystem has flown, fired, or been demonstrated. The opportunity is assembly, not invention.

Building block	Status, verified June 12, 2026
EMALS (US Navy)	Operational on USS Gerald R. Ford since 2017 incl. combat deployment; 484 MJ flywheel storage, 45 s recharge. CVN-79 (Kennedy) completed builder's trials Feb 2026, delivery ~Mar 2027. 2023 AFOSR-filed report recommends evolving EMALS for lunar launch.
Superconducting maglev	603 km/h crewed record (JR Central L0, 2015) still stands; Shanghai Transrapid commercial since 2003; Chūō Shinkansen now expected NET ~2034 after the Shizuoka settlement (early 2026).
REBCO superconductors	Tc ≈ 92 K. All-REBCO records: 26.86 T (ASIPP, 2024) → **28.2 T** (CHMFL, 2025); all-superconducting record **35.1 T** (ASIPP, Sept 2025); 45.5 T hybrid (NHMFL, 2019). CFS demonstrated 20 T SPARC-class coils (2021); SPARC assembly underway (first TF coil placed Jan 2026), first-plasma targeted 2026, Q>1 targeted 2027; ARC (Virginia) early 2030s. Jc > 1,000 A/mm² at 30 T *at 4.2 K*. Launch LSM needs only 5–10 T at the gap.
MHD plasma window	Invented by A. Hershcovitch (Brookhaven), patented 1995; holds ~9 atm; **demonstrated** for non-vacuum e-beam welding at cm-scale apertures. Scaling to the Ø4 m bore (2.8× baseline power) is the program's R&D centerpiece, Phase 1 tests it at full bore.
X-59 shock shaping	First flight Oct 28, 2025; **first supersonic flight June 5, 2026** (Mach 1.1); Mach 1.4 acoustic points next; XVS camera-vision cockpit flight-proven.
High-g electronics	Gun-launched guidance routinely qualified to 15,000–20,000 g (Excalibur-class). SELENE's 100 g cargo mode is conservative by two orders of magnitude.
EM launch-assist market	Auriga Space: $12.2M cumulative incl. AFWERX D2P2; hypersonic-test-as-revenue model proven in market; outdoor track (Thor) planned 2026.
Orbital manufacturing demand	Flawless Photonics: ~11.9 km ZBLAN drawn on ISS (Feb–Mar 2024; 1,141 m/day vs 25 m prior record); Luxembourg production expansion; published performance characterization still pending, flagged honestly.
Lunar mass-driver physics	O'Neill, NASA Ames studies 1976–77; Mass Driver One demo 1977. SJSU sizing study (2023): 25.4 kg pellets, 2.4 km/s, 10 s cadence, 8.7 MW polar solar, ~1.0 MA per bucket coil.
Lunar intent at scale	SpaceX lunar pivot + mass-driver statements (Feb–Mar 2026). Founder remarks, not a program of record, but the direction of the largest launch company on Earth.
Fusion pull (He-3 context)	NIF ignition Dec 2022 (3.15 MJ); best shot now **8.6 MJ, target gain >4** (Apr 2025), ignition repeated 9+ times. He-3 demand is already contracted, but for **quantum-cryogenics**, not fusion: Interlune–Bluefors offtake (Sept 2025), up to 10,000 L/yr from 2028, prospecting mission 2027. No commercial D-He3 reactor yet exists (see §13.4).

12. The Hard Parts

A document about a project this large that doesn't name the real problems is salesmanship, not engineering.

- **Plasma window scaling.** Centimeter demos → Ø4.0 m bore at 2.8× baseline power. First R&D dollar; if it doesn't scale, the muzzle needs a different answer and we must learn that at demonstrator scale, cheaply.
- **Cost per kilometer is unknown.** The $20–250/kg projection lives or dies on vacuum-rated superconducting track cost. Nobody has published a credible figure. Chimborazo Phase 1 produces the first real one.
- **The El Arenal piers.** 110–590 m A-frames in a 0.4 g seismic zone, beyond Millau's 343 m record. Partial trenching, rock-buttressed spans, or accepting a lower exit are the studied outs.
- **The reserve.** Chimborazo sits inside a protected fauna reserve under a rights-of-nature constitution. The legal path may be harder than the rock. Stated plainly because pretending otherwise would be fatal later.
- **SELENE's first bill is a rocket bill.** First coils, power, and radiators ride to the Moon at today's prices; the starter track must fit one or two heavy-lift flights.
- **Catching is harder than throwing.** Until the first cislunar catch happens, export revenue is theoretical. Wide-cone tugs first, tethers later.
- **Polar ice uncertainty.** Shackleton-specific reserves are unproven (bounded ≤5–10 wt% in surveyed regolith). The early business plan must close on regolith, oxygen, and metal even in the lean-ice case. Chang'e-7 data lands within a year.
- **The treaty question.** The Outer Space Treaty is silent on commercial mass drivers; a kinetic launcher is inherently dual-use. That ambiguity buys scrutiny on one side and AFWERX funding on the other. EMALS and GPS both walked this road.

13. The Lunar Export Economics, by the Numbers

The honest accounting: launching is the cheap part. Making material worth launching, and getting it home, are the walls. Every figure here is derived from first principles and cited.

A dedicated deep-research and fact-check pass (June 13, 2026; ~60 primary sources) re-derived SELENE's export case. It survives, but only when stated precisely. The corrections below make the architecture *stronger*, because they replace a fragile fusion bet with a robust cislunar-logistics business.

13.1 Launch energy is trivial, the physics floor

SELENE accelerates a pod to lunar escape velocity, 2.38 km/s. The energy is purely kinetic:

```
E_launch = ½v² = ½ × (2,380 m/s)² = 2.832 × 10⁶ J/kg = 0.787 kWh/kg (ideal)
Wall-plug at 84% drive efficiency: 0.787 / 0.84 = 0.94 kWh/kg
Cross-check (SJSU mass-driver design, Miller 2023): 8.7 MW ÷ 2.54 kg/s = 3.42 MJ/kg = 0.95 kWh/kg ✓
Electricity cost at $0.06/kWh: 0.94 × $0.06 ≈ $0.06 per kg to escape
```

So the launch itself costs pennies per kilogram and is *not* the economic constraint. This matches the O'Neill/NASA mass-driver heritage and the 2023 San José State sizing study, both built on 1,000 g, ~2.4 km/s. (O'Neill, NASA SP-428, 1977; Miller, SJSU AE thesis, 2023.)

13.2 The processing wall, the number that reframes everything

Energy to *escape* is trivial; energy to *refine* is not. Run the same 10 MW solar plant three ways:

```
Launch (raw mass): 10,000 kW ÷ 0.80 kWh/kg = 12,500 kg/h = ~300 t/day
Oxygen (H₂-reduction): 10,000 kW ÷ 24 kWh/kg = 417 kg/h = ~10 t/day (Lordos, PNAS 2025)
Solar-grade silicon: 10,000 kW ÷ 150 kWh/kg = 67 kg/h = ~1.6 t/day (Siemens-class)
⇒ refining costs 30× (O₂) to ~190× (Si) more energy per kg than launching it
```

> **This is the load-bearing insight.** "~300 t/day" is *raw or lightly-processed* mass throughput. Anything refined is gated by the processing plant, not the launcher, a solar-powered lunar industry is bottlenecked by ISRU energy, not launch energy. SELENE's job is to make the launch free; the power plant decides how much *finished* product exists. (Lordos et al., PNAS 122:e2306146122, 2025; Guerrero-Gonzalez & Zabel, Acta Astronautica 203:187, 2023.)

13.3 What is actually worth exporting

Commodity	Abundance / energy	Best customer	Verdict
Propellant (LOX/LH_2) from polar ice	ice ~5.6 wt% (LCROSS); ~tens kWh/kg	**Cislunar depots**	Robust near-term, sold in space
Oxygen from regolith	40–45 wt% O; ~24 kWh/kg LOX	Cislunar / life support	Strong, in-space
Structural metals (Fe/Al/Ti)	Fe 14–17%, Al 10–18%, Ti 5–8%	Lunar / cislunar build	In-space only (Moon beats asteroids for Ti, Al)
Helium-3	~4–10 ppb; ~5,000 GJ/kg to extract	Earth (quantum cryo)	Luxury commodity, kg/yr (§13.4)
Platinum-group metals	~1–5 g/t, meteoritic, unproven	Earth	Speculative, asteroids win

The peer-reviewed consensus is blunt: **cislunar use beats Earth-return for essentially everything** except the very highest value density. This does not weaken SELENE, it sharpens the "exporter from day one" claim to its defensible form: *SELENE sells propellant, oxygen and metal into cislunar space*, where a lunar kilogram is worth multiples of an Earth-lifted one. Earth-return is reserved for ultra-high-value goods. (Crawford, arXiv:1410.6865, 2015; Metzger, arXiv:2303.09011, 2023.)

13.4 Helium-3, honestly

The popular "Moon powers Earth with fusion fuel" story does not survive the arithmetic. The grade is the problem:

```
Grade 10 ppb ⇒ 10 mg He-3 per tonne ⇒ 10⁵ tonnes regolith per kg He-3 (100% yield)
At 80% thermal yield: ~1.25 × 10⁵ t mined & heated per kg
Heat fines to 700 °C: c_p ≈ 0.8 kJ/kg·K, ΔT = 670 K ⇒ 0.536 GJ per tonne
No heat recovery: 1.25×10⁵ t × 0.536 GJ/t ≈ 67,000 GJ/kg He-3 (≈ 18.6 GWh/kg)
With 85% recovery: ~10,000 GJ/kg; at 20 ppb + recovery: ~5,000 GJ/kg
Cross-check (NASA M-3 miner): 12.3 MW thermal, 33 kg/yr ⇒ 5,289 GJ/kg ✓ (within 5%)
```

That extraction energy, thousands of GJ per kg, is **~10^4× the launch energy**, so the crusher is lunar earth-moving and heat, never SELENE's gun. And the demand is misunderstood: the ~$20M/kg price is a **quantum-cryogenics** scarcity price (dilution-refrigerator coolant), not a fusion price. No commercial D-He3 reactor exists, D-He3 is harder to ignite than D-T, and the USGS (2022) lists lunar He-3 as "inferred unrecoverable." **Verdict: He-3 closes as a kilogram-per-year luxury commodity into quantum/medical markets (Interlune–Bluefors, up to 10,000 L/yr, 2028–2037), a real, high-margin niche, but it is not, today, fusion energy.** (Olson, NASA ASCEND 2021; Fegley & Swindle 1993; USGS 2022; Interlune–Bluefors, 2025.)

13.5 Lunar manufacture of hardware, what is real

Oxygen and crude metal are the mature part; finished electronics are not.

Process	Output	TRL	Basis
Carbothermal reduction (Sierra Space / NASA CaRD)	O_2	**6**	Thermal-vacuum, −45→1,800 °C, on simulant
H_2 reduction of ilmenite	O_2 + Fe	5	Breadboard; low total O_2 yield
Molten regolith electrolysis (MRE)	O_2 + Fe/Si/Al/Ti	3–4	Gram-scale lab, ambient, simulant
"Blue Alchemist" MRE → solar cells	PV chain (99.999% Si, Al wire)	~3	Company claim, lab on simulant
Vacuum-evaporated solar-grade Si	thin-film Si	2–3	<1 ppm impurities; *below* chip grade

Solar cells and aluminium wiring are plausible; **semiconductor (logic-chip) grade silicon is not demonstrated, and copper is essentially absent** on the Moon, so aluminium is the lunar conductor. That is not a compromise:

```
σ(Al) ≈ 61% IACS of Cu ⇒ equal-resistance Al needs 1/0.61 = 1.64× the cross-section
but ρ(Al)=2.70, ρ(Cu)=8.96 g/cm³ ⇒ equal-resistance Al mass = 1.64 × (2.70/8.96) = 0.49 × copper
⇒ an aluminium conductor is ~half the mass of copper for the same conductance, ideal where Al is abundant
```

(Blue Origin "Blue Alchemist," 2023; Sierra Space carbothermal, 2024; Ignatiev/NASA vacuum-Si; Azami et al., arXiv:2408.05823, 2024; Lunar Sourcebook Ch.8 for copper scarcity.)

13.6 Getting it home, re-entry dominates

```
A pod at 2.38 km/s leaves the lunar sphere of influence ≈ at rest, then falls down Earth's well.
Vis-viva, 384,400 km → 120 km entry interface: v_entry ≈ 10.99 km/s (reproduces Apollo/Orion ~11.1)
E_entry = ½ × (11,000)² = 60.5 MJ/kg = 21.4 × the launch energy (2.83 MJ/kg)
```

Re-entry, not launch, is the energy-management problem. Precedent sets the envelope: Stardust entered at 12.9 km/s (~1,200 W/cm^2, ~34 g); Apollo at ~11 km/s (~6–7 g guided); Hayabusa's heat shield was 53% of capsule mass. A ~500 kg cargo pod takes the simplest proven path, **ballistic entry + parachute** (Genesis/Stardust class): ~30–34 g (fine for hardware), peak ~600–1,000 W/cm^2, TPS ~15–35% (~100–175 kg). Orbital catch (a rotating skyhook) avoids the heat entirely but needs a facility ~200× the pod mass, worth it only at high cadence. (NASA NTRS: Stardust 20090004445, Apollo TN D-5399, Hayabusa 20120011648; HASTOL NIAC.)

13.7 The site checks out

SELENE's near-continuous-solar premise is confirmed by LOLA/LRO topographic illumination modelling: the Shackleton–de Gerlache Connecting Ridge receives **92.3% annual sunlight at 2 m height, 95.6% at 10 m**, with a longest continuous darkness of only 3–5 days. (Gläser et al., Icarus 243:78–90, 2014.)

Key sources (this section): Crawford, *Lunar Resources: A Review* (arXiv:1410.6865, 2015) · Metzger, *Economics of In-Space Industry* (arXiv:2303.09011, 2023) · Lordos et al., *PNAS* 122:e2306146122 (2025) · Guerrero-Gonzalez & Zabel, *Acta Astronautica* 203:187 (2023) · Olson, NASA ASCEND 2021 (He-3 mining) · Gläser et al., *Icarus* 243:78 (2014) · Colaprete et al., *Science* 330:463 (2010, LCROSS) · Miller, SJSU mass-driver thesis (2023) · Azami et al. (arXiv:2408.05823, 2024) · NASA NTRS entry-physics (Stardust, Apollo, Hayabusa) · Interlune–Bluefors offtake (2025). Full report: KERAUNOS lunar-export deep research, 2026-06-13.

14. What KERAUNOS Unlocks

Orbital manufacturing

- **ZBLAN fiber**, up to 100× lower theoretical loss than silica; ~12 km drawn on ISS in 2024; customers queue on freight prices. KERAUNOS is the freight price. (Performance proof still pending publication, the bet is stated, not oversold.)
- **Orbital bioprinting**, organs that collapse under their own weight in 1 g; daily return pods turn it from stunt to logistics business.
- **Debris salvage**, thousands of tonnes of aerospace alloy already in orbit becomes ore when tug-hours are cheap.

Geopolitical position

- **The Space Panama Canal**, the host nation stops paying for access and starts charging for it.
- **Presence as a supply chain**, 80 t/day secures orbital slots and lunar territory as a logistics fact, not a flag.
- **The deterrence dividend**, constellation replenishment in hours, not months.

Fusion economy

- **The He-3 supply chain, as a luxury commodity, not a fusion fantasy.** Be honest about what He-3 is and isn't (full derivation in §13.4). The ~$20M/kg price is a **quantum-cryogenics** scarcity price (dilution-refrigerator coolant, Interlune's actual contracted buyers, e.g. Bluefors, up to 10,000 L/yr 2028–2037), *not* a fusion-fuel price. There is no commercial D-He3 reactor; D-He3 is harder to ignite than D-T, and the USGS (2022) classes lunar He-3 "inferred unrecoverable." At ~4–10 ppb in regolith the extraction is enormous earth-moving (~10^5 tonnes mined and heated per kg), so the cost driver is lunar energy, not SELENE's launch. **The honest business closes at the kilogram-per-year scale into quantum/medical markets**, a real, high-margin niche, and only *becomes* civilization-scale fuel logistics if D-He3 fusion is ever solved. SELENE is positioned for that future without betting on it.
- **The energy loop.** Trackside storage doubles as grid-scale battery between launch windows; the same plant powers debris-sweeping lasers over the corridor. Nothing idles.

Verification note: all external claims in this revision were re-checked against primary reporting and literature on June 12, 2026; all physics re-derived from first principles the same day. Annex A (BRONTE Chimborazo Design Map) is bound after Section 4. The SELENE Design Map (KER-SEL-TRK-002) is issued as a companion sheet.